Nickel-Catalyzed Cross-Coupling Reactions of 4-Mesylcoumarins with Aryl Halides: Facile Synthesis of 4-Substituted Coumarins

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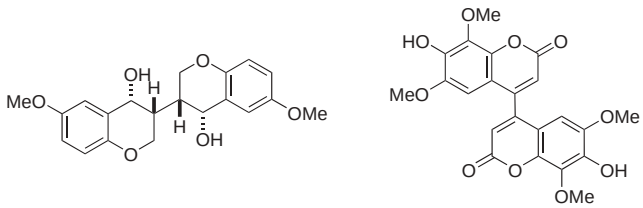
Abstract: A new and efficient nickel-catalyzed cross-coupling reaction between 4-mesylcoumarins and aryl halides was developed to give a number of 4-arylcoumarins in good yield under mild conditions. Unlike the previously reported coupling examples, this new method allows the direct cross-coupling of 4-mesylcoumarins with aryl- or vinyl halides in the $NiCl_2(PPh_3)_2/PPh_3/Zn$/toluene system. This has greatly facilitated the synthesis of biologically useful 4-substituted coumarins.

Key words: coumarin, aryl halide, nickel-catalyzed, homocoupling, cross-coupling

In recent years, coumarins have attracted a lot of attention because of their important physiological properties and their presence as important structural units in a large variety of biologically active natural products.[1,2] Interest in the synthesis of these compounds has stimulated the development of numerous methods for the construction of coumarin structures and their derivatives.[3–8] Among the various methods developed for the synthesis of 4-substituted coumarins, the cross-coupling approaches by using transition metal such as palladium or nickel catalyst are the most direct and attractive. Recently, remarkable progress has been made in this area.[4–8] For example, the palladium-catalyzed Stille-type cross-coupling of 4-trifylcoumarin or 4-tosylcoumarin with organostannanes,[5] and Suzuki-type cross-coupling of 4-trifylcoumarins or 4-halocoumarins with arylboronic acids[6] are found efficient for the preparation of 4-substituted coumarins. In addition, the nickel-catalyzed cross-coupling of 4-tosylcoumarins with terminal acetylenes and organozinc reagents,[7] 4-diethylphosphonoxycoumarins with organozinc reagents[8] are also successfully developed to synthesize various 4-substituted coumarins.

In our laboratory, we are interested in nickel-catalyzed coupling reactions and their application in natural products synthesis. In the synthesis of *bis*benzopyran-4-ol (**1**), a natural product isolated from the root of *Aloe barbadensis*,[9] we have developed a new reagent system [$NiCl_2(PPh_3)_2/PPh_3/Zn/NaH$/toluene] for the homocoupling of aryl halides or vinyl halides (Figure 1).[10] Using this modified Ullmann-type coupling reaction system, we have also successfully accomplished the total synthesis of a biologically active new *bis*coumarin compound, 4,4′-bi*iso*fraxdin (**2**),[11] by the homocoupling of its corresponding 4-tosylcoumarin.[12] In the continuation of our work, we wish to prepare a number of different 4-substituted coumarins. Combining the above two successes, we envisioned the potential of nickel-catalyzed cross-coupling of 4-sulphonylcoumarins with aryl halides to produce 4-aryl coumarins. It would be highly practical if these two readily available species could be directly coupled. Herein, we report our preliminary results of the cross-coupling reaction between 4-mesylcoumarins and aryl halides in the presence of nickel catalyst.



1 Bisbenzopyran-4-ol **2** 4,4'-Biisofraxidin

Figure 1

To chose the best 4-coumarin sulfonates for cross-coupling attempts, we first investigated the homocoupling reaction of 4-tosylcoumarin, 4-trifylcoumarin and 4-mesylcoumarin under our standard coupling conditions [$NiCl_2(PPh_3)_2/PPh_3/Zn/NaH$/toluene, 90 °C].[10] The experimental results are listed in Table 1. Among the three substrates (**3a**–**c**) tested, the 4-trifylcoumarin (**3b**) should have the highest reactivity in the coupling reaction, however, it only gave the homocoupling product in low yield (41%) along with 40% of the reduced product (entry 2).

Interestingly, the mesylate **3c** underwent homocoupling reaction smoothly and afforded the best yield of 75% after five hours (entry 3). Similar yield (76%) was achieved when a substituted 4-mesylcoumarin (**3d**) was employed. It seems likely that the 4-mesylcoumarin (**3c**) is an appropriate substrate for cross-coupling study. To our knowledge, examples of using aryl mesylates in nickel-catalyzed homocoupling and cross-coupling reactions has been successfully reported in the pioneering work of Percec.[13]

To begin attempting the nickel-catalyzed cross-coupling reaction of 4-mesylcoumarin (**3c**) with aryl halide, we first sought to find a best reaction condition that would minimize the homocoupling of both reactants. Thus, the

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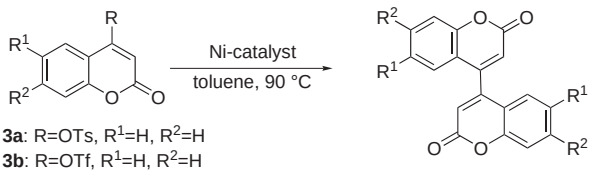
Table 1 Homocoupling of 4-Coumarin Sulfonates under the Standard Conditions[a]



3a: R=OTs, R^1=H, R^2=H
3b: R=OTf, R^1=H, R^2=H
3c: R=OMs, R^1=H, R^2=H
3d: R=OMs, R^1=Me, R^2=H

4

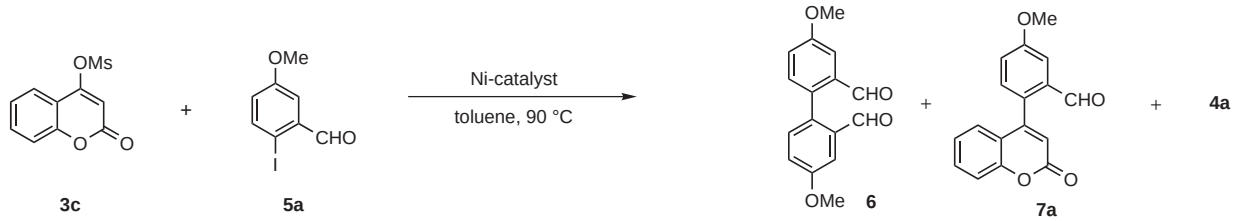
Entry	Substrate	Dimer	Time (h)	Yield (%) [b]
1	**3a**	**4a**	30	62
2	**3b**	**4a**	13	41[c]
3	**3c**	**4a**	5	75
4	**3d**	**4b**	18	76

[a] All reactions were performed using 0.2 equiv of $NiCl_2(PPh_3)_2$, 0.4 equiv of PPh_3, 0.1 equiv of NaI, 5 equiv of NaH, 3 equiv of Zn in 0.25 M of toluene at 90 °C.
[b] Isolated yield.
[c] Reduced product coumarin was observed in 40 % yield.

reaction of **3c** with **5a** was chosen and examined (Table 2). We found that the cross-coupling product **7a** could be formed under our reaction system, however, the yield was low (21%, entry 1). To avoid the homocoupling of the corresponding substrates, we added the solution of **5a** in toluene dropwise through a micro-syringe pump to the mixture of **3c** and catalyst in toluene. The slow addition of **5a** increased the yield of **7a** to 42% (entry 5). In addition, we observed a higher yield (58%) in the absence of Bu_4NI and NaH (entry 6). The bidentate ligand dppe had lower efficiency than the monodentate ligand PPh_3 (entry 7 vs. 6). Surprisingly, the use of benzene as reaction solvent caused a dramatic decrease in the yield of cross-coupling product and led to the formation of both homocoupling by-products (entry 8). Based on these results, we decided to choose the conditions in entry 6 as the best for further study.[14]

Gratifyingly, the catalytic cross-coupling reactions of **3c** with a series of aryl halides were successful to afford the corresponding 4-substituted coumarins in moderate to good yields (Table 3). As shown in Table 3, the mesylate **3c** reacted smoothly with both aryl bromide and iodide. In most cases, the iodide and bromide show similar reactivity, the reaction usually could be done in two to five hours. A little better yield was achieved by using the iodide. When aryl iodide **5e** was employed, the reaction was faster and completed in one hour to give a moderate yield (entry 5).[15] It is noteworthy that the reaction was not hampered by *ortho*-substituents. Unlike most other coupling reactions, it seems likely that the steric effect is not the main problem here. Furthermore, the functional group such as aldehyde is tolerated under the reaction conditions. Other than the aryl halides, the vinyl halide **5j** could also react with **3c** to produce the cross-coupling product **7f** in good yield (83%, entry 10).[16]

Table 2 Cross-Coupling between 4-Mesylcoumarin **3c** and Aryl Iodide **5a** under Different Reaction Conditions[a]



3c **5a** **6** **7a**

Entry	Bu_4NI (equiv)	NaH (equiv)	$Time_1$ (h)[b]	$Time_2$[c] (h)	Yield of **6** (%)	Yield of **7a** (%)	Yield of **4a** (%)
1	0.1	6	0	4	34	21	18
2	0.1	6	0.5	7	21	36	0
3[d]	0.1	6	1[c]	6	40	11	14
4	0.1	5	2	11	14	22	0
5	0.1	6	3	9	0	42	0
6			3	4	18	58	0
7[e]			3	5	74	20	0
8[f]			3	5	13	27	13

[a] All reactions were performed using 0.2 equiv of $NiCl_2(PPh_3)_2$, 0.4 equiv of PPh_3, 3 equiv of Zn in toluene at 90 °C.
[b] Time of the addition of substrate **5a**.
[c] Time after the addition of substrate **5a**.
[d] Compound **3c** was added slowly during 1 h.
[e] Instead of PPh_3, dppe was used.
[f] Benzene was used as solvent instead of toluene.

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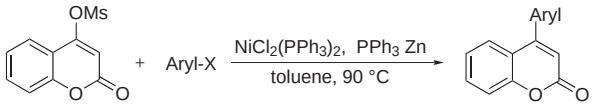
Table 3 Cross-Coupling of **3c** with Aryl Halides **5**[a]



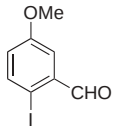
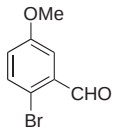
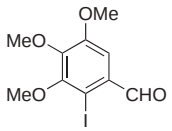
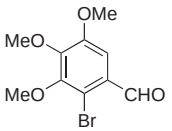
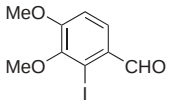
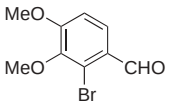
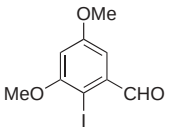
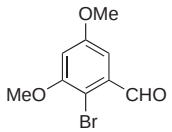
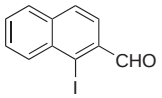
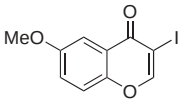
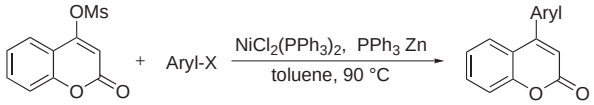
Entry	Aryl-X **5**	Product	Yield (%)[b]	Time (h)[c]
1	 **5a**	**7a**	58	2
2	 **5b**	**7a**	85	2
3	 **5c**	**7b**	71	2
4	 **5d**	**7b**	70	3
5	 **5e**	**7c**	72	1
6	 **5f**	**7c**	67	4
7	 **5g**	**7d**	67	2
8	 **5h**	**7d**	52	5
9	 **5i**	**7e**	63	5
10	 **5j**	**7f**	83	3

Table 3 Cross-Coupling of **3c** with Aryl Halides **5**[a] (continued)

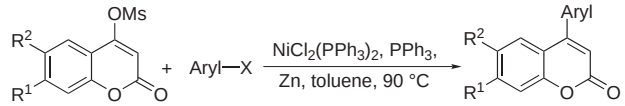


Entry	Aryl-X **5**	Product	Yield (%)[b]	Time (h)[c]

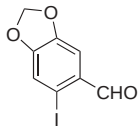
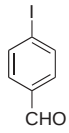
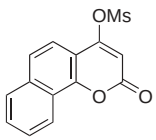
[a] Reaction conditions: a solution of **5** (0.6 mmol) in 3 mL of toluene was added during 3 h into the solution of **3c** (0.5 mmol), $NiCl_2(PPh_3)_2$ (0.2 equiv), PPh_3 (0.4 equiv), Zn (2–3 equiv) in 3 mL of toluene.
[b] Isolated yield based on mesylate **3c**.
[c] Reaction time after the addition of **5**.

We also investigated the possibility of the cross-coupling of diverse 4-mesylcoumarins **3d–g** with a variety of aryl or vinyl halides (Table 4). Substitution groups (such as methyl, fluoro, methoxy and so on) on the coumarin scaffold did not affect the coupling efficiency. Iodo- or bromo-substituted benzaldehydes were treated with those

Table 4 Cross-Coupling of Diverse 4-Mesylcoumarins with a Variety of Aryl or Vinyl Halides[a]



3d: R^1=H, R^2=Me
3e: R^1=H, R^2=F
3f: R^1=H, R^2=OMe
5 X=I, Br
7

Entry	**3**	Aryl-X **5**	**7**	Yield (%)[b]	Time (h)[c]
1	**3d**	**5c**	**7g**	69	4
2	**3d**	**5k**	**7h**	83	2
3	**3d**	**5a**	**7i**	53	5
4	**3e**	 **5m**	**7j**[18]	90	2
5	**3f**	**5h**	**7k**	90	5
6	**3f**	 **5n**	**7l**	55	5
7	 **3g**	**5n**	**7m**[19]	63	19

[a] Reaction conditions are the same as in Table 3.
[b] Isolated yield based on 4-mesylcoumarin s **3**.
[c] Reaction time after the addition of **5**.

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4-mesylcoumarins to obtain the corresponding coupling products in moderate to high yields (entries 1, 3–7). 3-Iodo-6-methoxy-benzopyran-4-one (**5k**) also reacted with mesylate **3d** to give 83% yield of the corresponding product **7h** (entry 2).[17]

In summary, we have demonstrated a new nickel-catalyzed cross-coupling of 4-mesylcoumarins with aryl or vinyl halides in the $NiCl_2(PPh_3)_2/PPh_3/Zn$/toluene reaction system. A number of useful 4-arylcoumarin compounds were obtained in good yields under mild conditions. Unlike the previously reported coupling examples, this new coupling method avoids the use of toxic organostannanes, sensitive organozinc reagents and expensive arylboronic acids; on the contrary, it allows the direct cross-coupling of easily accessible 4-mesylcoumarins with aryl or vinyl halides. Furthermore, the functional group such as aldehyde is tolerated under the reaction conditions. This has greatly facilitated the synthesis of a variety of biologically important 4-substituted coumarins. Application of this coupling method to construct a library of 4-coumarins is currently underway.

Acknowledgment

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(14) **General Procedure for the Synthesis of 7:** To an oven-dried Schlenk flask (25 mL) was charged 4-coumarin compound **3** (0.5 mmol), $NiCl_2(PPh_3)_2$ (0.2 equiv), PPh_3 (0.4 equiv), zinc dust (2.2 mmol) and dry toluene (3 mL) under argon. The resulting mixture was heated to 90 °C. The aryl halide **5** (0.6 mmol) was then added dropwise through micro-syringe pump over 3 h. After the addition, the reaction mixture was stirred at the same temperature until completion. The reaction mixture was cooled down to r.t., quenched with 5% HCl and diluted by CH_2Cl_2. The mixture was stirred vigorously at r.t. until the solution was clear. The organic layer was separated and the water layer was extracted with CH_2Cl_2 (3 × 10 mL). The combined organic layer was washed with sat. aq $NaHCO_3$, brine, and dried over Na_2SO_4. The solution was concentrated under vacuum; the obtained residue was purified by silica column chromatography to give the corresponding cross-coupling product **7** (selected characterization data of **7** are listed as follows).

(15) Compound **7c**. ^1H NMR (300 MHz, $CDCl_3$): δ = 9.69 (s, 1 H), 7.86 (m, 1 H), 7.54 (m, 1 H), 7.45 (m, 1 H), 7.22–7.16 (m, 2 H), 7.01 (m, 1 H), 6.40 (s, 1 H), 4.05 (s, 3 H), 3.70 (s, 3 H). MS (EI): *m/z* (%) = 310 (71) [M^+], 295 (100). IR (KBr): 3088, 2947, 2837, 2733, 1729, 1695, 1683, 1604, 1584, 1566 cm^{-1}. Anal. Calcd for $C_{18}H_{14}O_5$: C, 69.67; H, 4.55. Found: C, 69.86; H, 4.57.

(16) Compound **7f**. ^1H NMR (300 MHz, $CDCl_3$): δ = 8.08 (d, *J* = 2.1 Hz, 1 H), 7.64 (s, 1 H), 7.54 (d, *J* = 7.5 Hz, 1 H), 7.51 (d, *J* = 7.5 Hz, 1 H), 7.42–7.21 (m, 4 H), 6.45 (d, *J* = 2.1 Hz, 1 H), 3.93 (s, 3 H). ^{13}C NMR (75.5 MHz, $CDCl_3$): δ = 175.12, 160.67, 157.84, 154.08, 153.90, 151.45, 148.05, 132.41, 127.02, 124.91, 124.52, 120.90, 120.03, 118.95, 117.70, 117.45, 105.45, 56.28. MS (EI): *m/z* (%) = 320 (4) [M^+], 292 (10.00). IR (KBr): 3078, 2963, 1720, 1649, 1610, 1488, 1446, 1334, 1271 cm^{-1}. HRMS: *m/z* calcd for $C_{19}H_{12}O_5$: 320.0685; found: 320.0734.

(17) Compound **7h**. ^1H NMR (300 MHz, $CDCl_3$): δ = 8.05 (s, 1 H), 7.65 (s, 1 H), 7.55–7.30 (m, 4 H), 7.11 (s, 1 H), 6.42 (s, 1 H), 3.94 (s, 3 H), 2.34 (s, 3 H). ^{13}C NMR (75.5 MHz, $CDCl_3$): δ = 174.84, 160.58, 157.73, 153.66, 151.79, 151.20, 147.65, 133.95, 133.21, 126.31, 124.64, 120.69, 119.76, 118.46, 117.47, 116.90, 105.26, 59.01, 20.91. MS (EI):

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m/z (%) = 335 (18) [M^+ + 1], 334 (78) [M^+], 333 (21) [M^+ – 1], 306 (100). IR (KBr): 3078, 2959, 2929, 2498, 1719, 1698, 1646, 1618, 1568, 1488, 821 cm^{-1}. HRMS: *m/z* calcd for $C_{20}H_{14}O_5$: 334.0841; found: 334.0874.

(18) Compound **7j**. ^1H NMR (300 MHz, $CDCl_3$): δ = 9.70 (s, 1 H), 7.54–7.30 (m, 5 H), 6.80 (s, 1 H), 6.39 (d, *J* = 15 Hz, 2 H). MS (EI): *m/z* (%) = 312 (71) [M^+], 269 (100). IR (KBr): 3075, 2913, 2861, 1725, 1682, 1610, 1571, 1504, 1482, 1440, 1369, 1268, 1253 cm^{-1}. HRMS: *m/z* calcd for $C_{17}H_9FO_5$: 312.0434; found: 312.0385.

(19) Compound **7m**. ^1H NMR (300 MHz, $CDCl_3$): δ = 10.16 (s, 1 H), 8.09 (d, *J* = 9.0 Hz, 2 H), 7.89 (d, *J* = 9.0 Hz, 2 H), 7.74–7.58 (m, 4 H), 7.37 (d, *J* = 9.3 Hz, 1 H), 7.16 (d, *J* = 8.7 Hz, 1 H), 6.51 (s, CH, 1 H). MS (EI): *m/z* (%) = 301 (20) [M^+ + 1], 300 (91) [M^+], 272 (100). IR (KBr): 1733, 1631, 1601, 1549, 1500, 1470, 1375, 1209 cm^{-1}. HRMS: *m/z* calcd for $C_{20}H_{12}O_3$: 300.0786; found: 300.0809.

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